專題報告 1

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Legal Revisions Regarding Chemical Substance Management in Japan and Associated Changes in Exposure Assessment Methods: Current Status and Future Challenges
日本化學物質管理法律修訂及暴露評估方法相關變更：現況與未來挑戰

摘要： (中文)

2006 年前，日本化學物質管理體系對特定受管制物質（如有機溶劑及「指定化學物質」(約 120 種物質）施加極為詳細的雇主管理義務，而其他物質則幾乎未受管制。職場環境評估僅限於「職場環境測量」，一種區域監測方法。雖然政府於 2006 年引入要求雇主對管制清單外物質（超過 600 種物質）進行風險評估，但此措施效果有限。2012 年爆發膽管癌病例，造成大量受害者，主要由 1,2-二氯丙烷（一種未受管制物質）引起；隨後涉及其他職業性癌症的事件更凸顯此問題。受這些事件啟發，政府於 2016 年強制化學物質風險評估（涵蓋 670 種物質），隨後 對化學品管理系統進行重大改革，於 2024 年 4 月實施全面法律修正。

此新管理架構擴大了受強制風險評估的物質清單（目前涵蓋約 2,900 種物質）。該系統的一個關鍵面向是「自主管理」政策，將這些物質造成的意外或疾病後果歸責於雇主。此外，政府首次設立了「允許暴露限值」，目前涵蓋 178 種物質。此外，該系統要求所有處理化學物質的工作場所必須任命「化學物質作業主管」，並引入「化學物質管理專家」(如職業衛生師或職業衛生顧問）的角色，以提供整體化學品管理的指導。為了協助雇主進行風險評估，開發了一種名為「CREATE-SIMPLE」的簡化工具，現今已被廣泛使用。

同時，評估暴露的方法也經歷了重大變革。自 2021 年起，已引入以下測量方法：

(a) 使用個人取樣器的「呼吸區域監測方法」—測量工作環境的變化。
(b) 當工作場所在傳統工作環境測量中被認定為「不良環境」時，進行暴露測量以選擇適當的 RPE。
(c) 針對焊接工作的暴露測量 （注意：呼吸區濃度較高）。
(d) 針對具有指定允許暴露限值的物質進行個人暴露測量。
(e) 其他受風險評估物質的個人暴露測量。

關於（e）項，測量方法基本上由相關各方自行決定；實務上，通常遵循日本職業衛生學會於 2015 年發布的《個人暴露測量指引系列》。至於其他類別，(a) 至（d），政府對方法有相當詳細的規定，這使得整體暴露評估架構變得複雜。此外，由於對現行規範涵蓋物質進行職場環境測量的義務仍然存在，現行規範的特徵是「強制控制」與「自主管理」並存。

未來挑戰包括個人暴露測量的廣泛採用、 這些 方法的能力提升、 降低 測量/分析成本、風險評估專家的更廣泛可用性、支持中小企業（SMEs）以及衡量實務相關的倫理議題。

原文：Abstract

Prior to 2006, Japan’s chemical substance management system imposed extremely detailed obligations regarding specific regulated substances - such as organic solvents and "specified chemical substances" (totaling ab. 120 substances) - while leaving other substances virtually unregulated. Workplace environmental assessment was limited to "workplace environment measurement," an area monitoring method. Although the government introduced a requirement in 2006 for employers to make an effort to conduct risk assessments for substances outside the regulated list (over 600 substances), this measure proved largely ineffective. A turning point came in 2012 with an outbreak of bile duct cancer cases - resulting in many victims - primarily caused by 1,2-dichloropropane, an unregulated substance; subsequent incidents involving other occupational cancers further highlighted the issue. Prompted by these events, the government mandated risk assessments for chemical substances in 2016 (covering 670 substances) and then undertook a major reform of the chemical management system, resulting in the implementation of comprehensive legal amendments in April 2024.
This new management framework expanded the list of substances subject to mandatory risk assessment (currently covering ab. 2,900 substances). A key aspect of this system is the policy of "autonomous management," which places responsibility on employers for the consequences of accidents or illnesses caused by these substances. Additionally, the

government established - for the first time – “permissible exposure limits”, currently covering 178 substances. Furthermore, the system requires all workplaces handling chemical substances to appoint a "Chemical Substance Supervisor" and introduces the role of "Chemical Substance Management Expert" (such as occupational hygienists or occupational hygiene consultants) to provide guidance on overall chemical management. To support employers in conducting risk assessments, a simplified tool known as "CREATE-SIMPLE" was developed and is now widely used.
At the same time, methods for assessing exposure have also undergone significant changes. Since 2021, the following measurement methods have been introduced:

(a) A "breathing-area monitoring method" using personal samplers - a variation of workplace environment measurement.
(b) Exposure measurement conducted to select appropriate RPE’s when a workplace is deemed to be "poor environment" under conventional workplace environment measurements.
(c) Exposure measurement targeting welding works (note: high concentrations in the breathing zone).
(d) Personal exposure measurement for substances with designated permissible exposure limits.
(e) Personal exposure measurement for other substances subject to risk assessment.

Regarding (e), the measurement method is essentially left to the discretion of the parties involved; in practice, it often follows the "Guidelines for Personal Exposure Measurement" issued by the Japan Society for Occupational Health in 2015. For the other categories, (a) to (d), the government specifies methods in relatively fine detail, a situation that has complicated the overall framework for exposure assessment. Furthermore, as the obligation to conduct workplace environment measurements for substances covered by existing regulations remains in place, the current regulation is characterized by the coexistence of "mandatory control" and "autonomous management."

Future challenges include the widespread adoption of personal exposure measurement, competency in these methods, the reduction of measurement/analysis costs, the broader availability of experts consulting on risk assessment, support for small and medium-sized enterprises (SMEs), and ethical issues surrounding measurement practices.